

What to build next for the coach

A survey of what badminton coaches actually need, what the research says decides matches, and which of it this codebase is already most of the way to answering.

Date	28 August 2026
Status	Assessment, pending scope decision
Follows	The 2026-08-26 competitor review, and the video-attach work shipped today
Scope	15 candidate capabilities, each with a verdict. No implementation plan yet.

1 The short version

Every competitor surveyed has *either* automatic video analysis *or* a scorekeeper. As of today this app has both, plus court homography, pose and shuttle tracking. That combination is a category nobody currently occupies, and it is worth building around rather than diversifying away from.

The bottleneck the research keeps naming is that match analysis is done by hand. A human-authored point sequence is exactly the ground truth an automatic detector has never had, and an automatic detector is exactly the labour a note-taking coach has never had. Neither half is worth much alone. Bound together they are the product.

So the recommendation is not to add features outward. It is to **finish joining the two halves**, then make the match record answer the three questions coaches already answer with a clipboard.

THE HEADLINE

Four pieces of work, in order: **reconcile the points to the rallies, give players a real identity**, then the **error ledger** and the **serve ledger**. The sequence compounds, and none of it needs any new machine learning.

2 What the research says decides matches

Three findings shaped the recommendations more than anything else.

36 / 23 / 41

Winners, forced errors, unforced errors as a percentage split. The primary badminton statistic, from Rio 2016 analysis.

64.8%

Of rally-deciding moments were caused by a *bad* shot, not a good one. Good shots accounted for 35.2%.

84.3%

Of those bad shots converted into a lost point, where good shots converted at 69 to 73%.

Errors decide matches more than winners do

A 2025 perturbation study concluded that training should be reframed "from how to win points toward how to prevent losing them". Poor lobs and clears generated the costliest mistakes. This is a presentation decision as much as a data one: a match summary that **leads with errors rather than winners** is better coaching, not just a different chart.

Rally length is a diagnostic, not trivia

Elite rallies run 9 to 10 seconds and 7.5 to 9 strokes. A club rally averaging four strokes is a statement about errors, and a coach reads it instantly without being taught what to look for.

Club coaches are not elite coaches

Harvard's VIRD study, which found that analysis "relies heavily on manual note-taking due to lack of automatic data collection", studied Olympic and national staff who wanted flexible viewpoints on spatial data. Club-level video analysis is meanwhile sold as hours of coach time, around GBP 50 for a single match.

The club coach's scarce resource is **time**, not spatial fidelity. That argues for a match report he can read in a minute, and against an immersive analysis surface.

3 What the codebase already has

Claims below marked **VERIFIED** were read in this repository. Claims marked **REPORTED** come from the Python pipeline repo and are quoted from the earlier review rather than re-read.

- **VERIFIED** The seeded label set for every new account is exactly `Good shot` , `Forced error` , `Unforced error` . That is the same three-way division the literature uses as its primary statistic. Worth building on rather than around.
- **VERIFIED** The scoring fold already records more than any screen reads. Every point carries its game, who won it, the score before and after, its tags, its comment, and **who served it**. The server is a stored field, not a derivation.
- **VERIFIED** A match is created before any video exists, scored courtside, and can now take a video afterwards that runs through the clipping pipeline onto the same match page.
- **REPORTED** The rally detector already counts shuttle direction reversals to find rallies, and therefore already knows how many strokes each rally contained. Nothing persists it.

And the gaps, all verified:

- **No player identity**. Players are free text per match. Two matches cannot know they share a player, so no statistic can outlive one match.
- **No side on a tag**. A rally can be marked "Unforced error" without recording whose it was.
- **No shot count or placement on a clip**. The clip knows its timings and nothing about what happened inside it.
- **No point-to-rally mapping**. Deliberately deferred, and now the thing standing in front of everything else.

4 Capability verdicts

Fifteen candidates. The declines are listed rather than silently skipped, so the reasoning is on the record.

#	CAPABILITY	VERDICT	COST	WHY
1	Reconcile: map point N to rally N	BUILD FIRST	M	Already designed in full. Until it exists the points and the clips are two lists on one page. After it, every clip carries its score situation, and every capability below gets better.
2	Player identity	BUILD	S	Cheap, dull, longest tail. The precondition for every longitudinal question. Turns the product from <i>analyse this match</i> into <i>coach this player</i> .
3	Error ledger: winners, forced, unforced, per side	BUILD	S	The labels already exist and are already the right three. After #1 the side is derived rather than asked for: an error on a point the opponent won is this player's error.
4	Serve and receive outcomes	BUILD	XS	Who served is already a stored field. Points won serving versus receiving is a pure function of the existing log. Needs no video , so it lands on every match ever scored.
5	Rally length and stroke count	BUILD	S	The detector computes the stroke count and discards it. One column, one pipeline change. Cross-repo.
6	Sessions: matches grouped by date	BUILD REDUCED	S	The 20% of tournament mode that buys 80%. Club nights are many short matches on one evening. Not brackets, not seeding, not roles.
7	Court coverage heatmap, distance covered	BUILD LATER	M	Homography and pose already produce court coordinates, so the data exists. All the work is in the Python pipeline. Most-advertised feature in the category.
8	Shuttle landing placement chart	BUILD LATER	M	Where rallies ended, as a court plot. Needs trajectory endpoint extraction. Pairs naturally with #7.
9	Shot classification	LATER, REDUCED	L	What every competitor leads with, and the largest lift here. The research suggests most of the coaching value sits in a coarse attacking / neutral / defensive split. Do not start here.
10	Longitudinal trends across matches	BUILD LATER	M	Falls out of #2 plus #3 and #4. Worthless before #2, nearly free after it.
11	Technique scoring, per-shot form critique	DECLINE	L	The category is full of it, and it needs the most machine learning for the least defensible accuracy. Our edge is the match record, not the swing.
12	Drill library, session planning	DECLINE	M	A different product with entrenched incumbents. Nothing we know about a match makes us better at prescribing a drill.
13	Club admin: scheduling, payments, registration	DECLINE	L	A genuine pain, and genuinely a business-operations product. Building it would double the surface area and halve the focus.

#	CAPABILITY	VERDICT	COST	WHY
14	Tournament brackets, seeding, roles	DECLINE	L	Already declined in the 2026-08-26 review and the reasoning holds. #6 is the part worth taking.
15	Multi-sport	DECLINE	L	Already declined. The whole pipeline is badminton-specific.

5 The running order

1 Reconcile the points to the rallies

It is already designed, it is the promise the video-attach work implicitly made, and it is the multiplier. The error ledger needs it for side attribution. The heatmaps become filterable by score situation because of it. And the point count it produces is the first honest measure of how far the detector drifts.

2 Give players a real identity

Cheap and unglamorous, with the longest tail of anything here. Every statistic built before it is stranded inside a single match. Nothing longitudinal is possible until it exists.

3 The error ledger and the serve ledger

Together they are a match report a coach can act on, drawn entirely from data already in hand. The serve half in particular costs almost nothing and needs no video, which means it lands for every match ever scored rather than only the filmed ones.

4 Then stop and reassess

Rally length is a small cross-repo follow-up. The heatmaps, placement charts and shot classification are a second, larger programme that should get its own design once the first three have been in a coach's hands. What he reaches for will reorder them.

6 Constraints worth knowing before planning

- **Three of these are cross-repo.** Rally length, heatmaps and placement are Python pipeline work plus migrations against a table the pipeline writes. Any new column the app must read needs its own explicit grant, and a service-role smoke test will not catch its absence.
- **Player identity changes an existing shape.** Player names are a text array with a constraint today. A players table means a migration that keeps existing free-text names working, not a replacement.
- **The error ledger really wants reconcile first.** Built before it, it would have to ask the coach which side erred, adding a tap per rally to the one surface where taps are expensive. Built after, it is derived.
- **The score_logs migration is still unapplied on the server.** Every capability here reads or writes that table.

7 Three questions I could not answer from the code

- **Individuals or a club?** Player identity and longitudinal trends are worth far more to a coach following eight named players over a season than to someone recording their own matches.
- **During the match, or after it?** The error and serve ledgers are computable live, on the board, with no video. If a between-games readout is what is wanted, that is a different and smaller surface than a post-match page.
- **How far does the detector actually drift?** Nobody knows yet. The first reconciled match answers it, and the answer should shape the reconcile screen rather than be guessed at now.

8 Sources

- Long rallies and next rally performances in elite men's and women's badminton — PLOS One
- Perturbations in badminton: why avoiding bad shots is more important than executing good ones, 2025 — Sage
- Statistical differences in set analysis in badminton at the Rio 2016 Olympic Games — Frontiers in Psychology
- Match-play data according to playing categories in badminton: a systematic review — Frontiers in Sports and Active Living
- VIRD: immersive match video analysis for high-performance badminton coaching — Harvard VCG / IEEE TVCG
- CoachAI: a project for microscopic badminton match data collection and tactical analysis
- Nacsport: using video analysis to improve badminton performance
- goSmash and BadPro+, surveyed as current competitors

The full assessment, with file-level references for every verified claim, is committed at `docs/plans/2026-08-28-coaching-capability-review-design.md`.